

Hyperbolic Conservation Laws: Equation Systems

May 16, 2025

Introduction

This document presents the mathematical formulations of three hyperbolic conservation law systems: the Isentropic Gas System, the Shallow Water System, and the Euler Equation System. Each system is described in conservative form, $\partial_t \mathbf{U} + \partial_x \mathbf{F}(\mathbf{U}) = 0$, with conservative variables $\mathbf{U}$, primitive variables $\mathbf{W}$, fluxes $\mathbf{F}(\mathbf{U})$, and sound speeds.

1 Isentropic Gas System

1.1 Conservative Variables

$$\mathbf{U} = \begin{bmatrix} \rho \\ m \end{bmatrix} = \begin{bmatrix} \rho \\ \rho u \end{bmatrix}$$

- ρ : Density
- $m = \rho u$: Momentum
- u : Velocity (primitive variable)

1.2 Primitive Variables

$$\mathbf{W} = \begin{bmatrix} \rho \\ u \end{bmatrix}, \quad \mathbf{U} = \begin{bmatrix} \rho \\ \rho u \end{bmatrix}, \quad \mathbf{W} = \begin{bmatrix} \rho \\ \frac{m}{\rho} \end{bmatrix}$$

1.3 Flux

$$\mathbf{F}(\mathbf{U}) = \begin{bmatrix} \rho u \\ \rho u^2 + p \end{bmatrix}, \quad p = k\rho^\gamma$$

- p : Pressure, with k (gas constant, default: 1.0) and γ (specific heat ratio, default: 1.4)

1.4 Conservation Law

$$\partial_t \begin{bmatrix} \rho \\ \rho u \end{bmatrix} + \partial_x \begin{bmatrix} \rho u \\ \rho u^2 + k\rho^\gamma \end{bmatrix} = 0$$

1.5 Sound Speed

$$c = \sqrt{\gamma k \rho^{\gamma-1}}$$

2 Shallow Water System

2.1 Conservative Variables

$$\mathbf{U} = \begin{bmatrix} h \\ m \end{bmatrix} = \begin{bmatrix} h \\ hu \end{bmatrix}$$

- h : Water height
- $m = hu$: Momentum
- u : Velocity (primitive variable)

2.2 Primitive Variables

$$\mathbf{W} = \begin{bmatrix} h \\ u \end{bmatrix}, \quad \mathbf{U} = \begin{bmatrix} h \\ hu \end{bmatrix}, \quad \mathbf{W} = \begin{bmatrix} h \\ \frac{m}{h} \end{bmatrix}$$

2.3 Flux

$$\mathbf{F}(\mathbf{U}) = \begin{bmatrix} hu \\ hu^2 + \frac{1}{2}gh^2 \end{bmatrix}$$

- g : Gravitational acceleration (default: 9.81 m/s²)
- $\frac{1}{2}gh^2$: Hydrostatic pressure term

2.4 Conservation Law

$$\partial_t \begin{bmatrix} h \\ hu \end{bmatrix} + \partial_x \begin{bmatrix} hu \\ hu^2 + \frac{1}{2}gh^2 \end{bmatrix} = 0$$

2.5 Sound Speed

$$c = \sqrt{gh}$$

3 Euler Equation System

3.1 Conservative Variables

$$\mathbf{U} = \begin{bmatrix} \rho \\ m \\ E \end{bmatrix} = \begin{bmatrix} \rho \\ \rho u \\ E \end{bmatrix}$$

- ρ : Density
- $m = \rho u$: Momentum
- E : Total energy per unit volume
- u : Velocity (primitive variable)
- p : Pressure (primitive variable)

3.2 Primitive Variables

$$\mathbf{W} = \begin{bmatrix} \rho \\ u \\ p \end{bmatrix}, \quad \mathbf{U} = \begin{bmatrix} \rho \\ \rho u \\ \frac{p}{\gamma-1} + \frac{1}{2}\rho u^2 \end{bmatrix}, \quad \mathbf{W} = \begin{bmatrix} \rho \\ \frac{m}{\rho} \\ (\gamma-1) \left(E - \frac{1}{2} \frac{m^2}{\rho} \right) \end{bmatrix}$$

- γ : Specific heat ratio (default: 1.4)

3.3 Flux

$$\mathbf{F}(\mathbf{U}) = \begin{bmatrix} \rho u \\ \rho u^2 + p \\ u(E + p) \end{bmatrix}$$

3.4 Conservation Law

$$\partial_t \begin{bmatrix} \rho \\ \rho u \\ E \end{bmatrix} + \partial_x \begin{bmatrix} \rho u \\ \rho u^2 + p \\ u(E + p) \end{bmatrix} = 0$$

3.5 Sound Speed

$$c = \sqrt{\frac{\gamma p}{\rho}}$$

Notes

- These formulations correspond to the implementations in `isentropic_gas.py`, `shallow_water.py`, and `euler.py`.
- Numerical safeguards ensure $\rho, h, p > 10^{-10}$ to prevent division by zero or negative values.
- The systems are solved using Godunov-type methods with fluxes (e.g., HLLC, Roe), reconstructions (e.g., WENO5), and boundary conditions (periodic, reflective, transmissive).